



2 Real Analysis 2023

Tutorial 9

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The topology of $\mathbb{R}$

Problem 34 Decide whether the following sets are open, closed, or neither. Give a justification.

- (a) $\mathbb{N}$,
- (b) $\mathbb{Q}$,
- (c) $B = \{x \in \mathbb{R} : x \neq 5\}$,
- (d) $C = \{x \in \mathbb{R} : 1 \leq |x| \leq 2\}$,
- (e) $D = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$.

Problem 35 Consider the set $S = \{0\} \cup \left\{\frac{1}{n} : n \in \mathbb{N}\right\} \cup [3, 4) \cup \{5\}$. Find

- (a) all its limit points,
- (b) all its isolated points,
- (c) its closure $\overline{S}$.

Problem 36 Let $A, B \subseteq \mathbb{R}$. Prove the following properties of the closures.

- (a) $A \subseteq B \implies \overline{A} \subseteq \overline{B}$,
- (b) $\overline{\overline{A}} = \overline{A}$,
- (c) $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

Problem 37 Let $A \subseteq \mathbb{R}$. A point $x \in \mathbb{R}$ is called an *adherent point* of A , if any ε -neighborhood $(x - \varepsilon, x + \varepsilon)$ contains a point of A .

- (a) Prove that the closure $\overline{A}$ of the set A is exactly the set of all adherent points of A .
- (b) Prove that

$$\begin{aligned} & \{a \in \mathbb{R} : a \text{ is an adherent point of } A\} \\ &= \{a \in \mathbb{R} : a \text{ is a limit point of } A\} \cup \{a \in \mathbb{R} : a \text{ is an isolated point of } A\}. \end{aligned}$$

Problem 38 Let $A \subseteq \mathbb{R}$. A point $x \in \mathbb{R}$ is called a *boundary point* of A , if any ε -neighborhood $(x - \varepsilon, x + \varepsilon)$ contains a point of A and a point of its complement $\mathbb{R} \setminus A$. The set δA of all boundary points of A is called the *boundary* of A .

- (a) Prove that $\delta A = \delta(\mathbb{R} \setminus A)$.
- (b) Determine the boundaries of the following sets: $B = \{x \in \mathbb{R} : x \neq 5\}$, $C = \{x \in \mathbb{R} : 1 \leq |x| \leq 2\}$, $D = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$, $M = (0, 1] \cup \{2\} \cup [3, 4]$.